

SONIA REILLY

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EDUCATION

Courant Institute of Mathematical Sciences, New York University Ph.D. in Mathematics Thesis Advisor: Georg Stadler Supported by Department of Energy Computational Science Graduate Fellowship	New York, NY <i>Expected 2027</i>
Massachusetts Institute of Technology S.B. in Mathematics with Computer Science	Cambridge, MA <i>2021</i>

RESEARCH INTERESTS

Bayesian inverse problems, PDE-governed inverse problems, uncertainty quantification

RESEARCH EXPERIENCE

Courant Institute of Mathematical Sciences <i>Graduate Researcher, advised by Georg Stadler</i> <i>Amortized Methods for Hierarchical Bayesian Inverse Problems</i> • Developed scalable deterministic methods of marginalizing hyperparameters in linear Bayesian PDE inverse problems with Gaussian priors. • Extended state-of-the-art low-rank approximation methods to enable fast computation in problems with nonlinear prior hyperparameters, such as correlation lengths. • Demonstrated speed and accuracy on a time-dependent 3D advection-diffusion benchmark problem. <i>MCMC Sampling for Infinite-Dimensional Bayesian Inverse Problems</i> • Implemented a novel MCMC sampler designed for posterior distributions that emerge from infinite-dimensional Bayesian PDE inverse problems. • Combined elements of affine invariant methods and the preconditioned Crank Nicolson method to form a gradient-free ensemble sampler that targets high-dimensional, concentrated posteriors. • Tested the method against existing MCMC samplers on Gaussian regression problems of varying complexity, using NYU's Greene supercomputer to compute MCMC chains in parallel.	New York, NY <i>2022 – present</i>
Lawrence Berkeley National Laboratory <i>Graduate Research Intern, advised by Jeff Donatelli in James Sethian's group</i> • Designed a method to incorporate measurement uncertainty into the Multi-Tiered Iterative Phasing (M-TIP) algorithm for recovering the density of a molecule imaged using fluctuation X-ray scattering. • Successfully retrieved densities from images with added constant and variable Gaussian noise, and made progress towards adapting the algorithm to allow for Poisson-distributed data.	Berkeley, CA <i>Summer 2023</i>
Johns Hopkins University Applied Physics Laboratory <i>Postgraduate Researcher, advised by Sam Yee</i> • Tackled the tomographic inversion problem of recovering a map of atmospheric brightness from satellite data which records integrated brightness values. • Developed and implemented a numerical method using 2D piecewise linear basis functions and a regularized matrix solve, which is more generalizable to 2D and 3D than classical 1D methods.	Laurel, MD <i>Summer 2021</i>

Undergraduate Researcher, advised by William Swartz

Sep 2016 - Aug 2017, Summer 2018

- Designed and implemented a method of extracting high-resolution maps of Earth's outgoing radiation from low-resolution satellite data using spherical harmonics approximation and compressive sensing.
- Applied the method to extract spatial data from RAVAN, NASA's first earth science CubeSat mission, and co-authored a resulting publication in *Remote Sensing* in April 2019.

Massachusetts Institute of Technology

Cambridge, MA

Undergraduate Researcher, advised by Jörn Dunkel

Feb 2020 – Dec 2020

- Developed network topologies optimized to cancel out fluctuations in coupled oscillator systems governed by the swing equation, such as power grid models.
- Categorized the effect of the noise spectrum of the fluctuations on the optimal network structure.
- Quantified the resistance of these optimized networks to cascades of line failures in a power grid model.

Undergraduate Researcher, advised by Sai Ravela

Summer 2019

- Investigated the mathematics of deep learning by modeling the evolution of the weights of neural networks in stochastic gradient descent as a Fokker-Planck system.

PRESENTATIONS AND POSTERS

6. SIAM Annual Meeting 2026, Cleveland, OH, July 2026 (invited minisymposium talk)
Session: From Theory to Computation: Modern Methods for PDE Inverse Problems
Title: "Amortized Low-Rank Approximation for Hyperparameter Marginalization in PDE-Governed Bayesian Inverse Problems"
5. ICERM Workshop on Bayesian Inverse Problems and UQ, Providence, RI, March 2026 (poster)
Title: "Bayesian Inference for Latent Gaussian Models Governed by PDEs"
4. IMSI Workshop on Data Assimilation and Inverse Problems for Digital Twins, Chicago, IL, October 2025 (poster)
Title: "Bayesian Inference for Latent Gaussian Models Governed by PDEs"
3. DOE CSGF Annual Program Review Outgoing Fellow Presentation, Washington D.C., July 2025
Title: "Bayesian Inversion of Hierarchical PDE-Governed Problems Using INLA"
2. SIAM Conference on Computational Science and Engineering, Fort Worth TX, March 2025 (invited minisymposium talk)
Session: Investigating Inverse Problems using Bayesian Inference: Challenges and Advances
Title: "Bayesian Inversion of PDE-Governed Problems Using Integrated Nested Laplace Approximations"
1. Mid-Atlantic Numerical Analysis Day, Temple University, Philadelphia, PA, November 2024
Title: "Bayesian Inversion of PDE-Governed Problems Using Integrated Nested Laplace Approximations"

PUBLICATIONS

2. **S. Reilly**, G. Stadler. Amortized Low-Rank Approximation for Hyperparameter Marginalization in PDE-Governed Bayesian Inverse Problems. arXiv preprint: 2607.03355.
1. W. Swartz, S. Lorentz, S. Papadakis, P. Huang, A. Smith, D. Deglau, Y. Yu, **S. Reilly**, N. Reilly, and D. Anderson. "RAVAN: CubeSat Demonstration for Multi-Point Earth Radiation Budget Measurements". *Remote Sensing* 11.7 (Apr. 2019), 796. DOI: 10.3390/rs11070796.

TEACHING AND MENTORSHIP

Graduate Teaching Assistant, Courant Institute, NYU

MATH-UA 334: Mathematical Statistics

Spring 2025, 2026 and Fall 2024, 2025

- Led two weekly 75 minute recitation sections per semester, each for 40 undergraduate juniors and seniors, and held weekly office hours.
- Wrote example problems and solutions for recitation worksheets, prepared and posted reviews of lecture material, led exam review sessions, and engaged with students on the class discussion board.
- Student evaluations: 4.67/5 Spring 2026, 4.88/5 Fall 2025, 5.00/5 Spring 2025, 4.50/5 Fall 2024

MATH-UA 253: Linear and Nonlinear Optimization

Spring 2024

- Led weekly 75 minute recitation sections and held office hours for a class of 45 undergraduate seniors.
- Developed Julia coding tutorials, wrote example problems and solutions for weekly worksheets, and prepared reviews of lecture material. Graded exams and responded to discussion board questions.
- Delivered a 90-minute lecture as a substitute for the course instructor.
- Student evaluations: 4.75/5

Student Research Mentor, Courant Institute, NYU

- Anchi Lin, undergraduate student, Summer 2026

AWARDS

SIAM Student Travel Award

2026

- \$650 in travel funding to SIAM's Annual Meeting in Cleveland, OH

SIAM Student Travel Award

2025

- \$650 in travel funding to SIAM's Computational Science and Engineering conference

Department of Energy Computational Science Graduate Fellowship

2021-2025

- 4 years of \$45,000 per year in stipend, plus tuition and fees

NSF Graduate Research Fellowship

2021

- 3 years of stipend, tuition, and fees. (Awarded, but declined in favor of DOE CSGF.)

Henry M. MacCracken Fellowship, New York University

2021

- 5 years of PhD support, with no teaching requirement.

Johns Hopkins University Applied Physics Laboratory ASPIRE Scholarship

2017

- \$2,000 in college tuition scholarship, awarded for the best presentations by high school interns.

Research Science Institute, Massachusetts Institute of Technology

2016

- Competitive, tuition-free summer research program for high school seniors.

SERVICE AND OUTREACH

CAS-GSAS Student Network Mentoring Program, Graduate Mentor

2026

- Mentored an NYU rising senior applying to graduate programs in math.

Courant Student Organization, President

2023–2024

- Co-led the mathematics PhD student organization, including interfacing with the department, raising funds, and organizing visiting students' day.
- Oversaw a revamp of the student-organized website that hosts qualifying exam prep resources for incoming PhD students.

Diversity, Equity, and Inclusion Reading and Outreach Group, Member

2021–2024

- Led and participated in discussions on diversity and equity in mathematics and graduate education.

- Volunteered at a local high school, assisting in evaluating student math projects.

Association for Women in Math, Courant Organizing Committee Member 2021–2023

- Organized social events for women in the Courant math department, and led an effort to invite more women speakers to department colloquia.

Courant Graduate Student/Postdoc Seminar, Co-organizer 2021–2022

- Invited speakers and organized logistics for a seminar connecting first-year students with potential advisors.

MIT Undergraduate Society of Women in Math, Peer Mentor 2020–2021

- Mentored a younger undergraduate woman in navigating the math department at MIT.

North American Computational Linguistics Open Competition, Organizing Committee Member and Problem Writer 2017–2021

- Helped organize, contributed problems, served on the jury, and hosted the MIT location of this contest for high school students.

SKILLS

Software	Python, Julia, MATLAB, C/C++, CUDA, OpenMP
Languages	Spanish (fluent); German, Italian, French (proficient in reading)